

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	DEEP
Model	qwen3:8b

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Editorial Decision

MAJOR REVISION

Weighted Score: 3.4 / 5.0
Confidence: 83%

Executive Summary

The manuscript presents a novel physics-informed EANN framework for structural parameter identification using accelerometers under multi-objective optimization. It includes experimental validation and quantitative results, but requires significant improvements in structure, reproducibility, and novelty demonstration.

Decision Rationale

The manuscript demonstrates significant technical contribution and experimental validation, but requires major revisions to address critical issues in structure, reproducibility, and novelty demonstration. The proposed framework is promising, but the lack of literature review, detailed descriptions of figures and equations, and insufficient comparison with prior work prevent it from meeting IEEE standards for publication.

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	5.0	Excellent	■■■■■
Manuscript Structure	2.0	Weak	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	4.2	Good	■■■■■
Discussion & Conclusions	3.2	Acceptable	■■■■■
Figures & Tables	2.5	Weak	■■■■■
Scientific Writing	3.0	Acceptable	■■■■■

References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	1.5	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 5.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] The title is specific, informative, and clearly reflects the contribution of the work, which is the development of a physics-informed EANN framework for structural parameter identification using accelerometers under multi-objective optimization.
- [ABSTRACT] The abstract follows a structured format with clear background, objective, methods, results, and conclusions. It quantifies results (e.g., $CV \approx 0.07-0.10$) and avoids vague language.
- [KEYWORDS] The keywords are comprehensive, include domain-specific terms, and are not redundant with the title. They enhance discoverability and align with the content.

Minor Comments:

- [TITLE] The title is slightly long but remains informative and specific. It could be slightly shortened for improved readability without losing clarity.

Recommendations:

- [TITLE] Consider shortening the title slightly to improve readability while maintaining specificity, e.g., 'Physics-Informed EANN for Structural Parameter Identification via Accelerometer Data under Multi-Objective Optimization'.

StructureReviewer

Manuscript structure and logical coherence

Score: 2.0 / 5.0 | Confidence: 95%

Strengths:

- Clear title and keywords
- Experimental validation of the proposed framework
- Reproducibility of methodology described

Weaknesses:

- Missing literature review section
- Research problem not clearly stated in introduction
- Research question or hypothesis not explicitly declared
- Section VI and VII are duplicates of the discussion and conclusions
- Abstract claims not fully supported by results
- Keywords not aligned with content
- Figures and tables lack complete captions
- Equations not numbered or defined
- Statistical validity not reported
- Reproducibility not ensured due to lack of algorithmic detail
- No ethical compliance or conflict-of-interest declaration

Major Comments:

- The manuscript is missing a literature review section, which is critical for establishing the context and novelty of the work.
- The research problem is not clearly stated in the introduction, making it difficult to assess the significance of the work.
- The research question or hypothesis is not explicitly declared, which undermines the clarity of the study's objectives.

- Sections VI and VII are duplicates of the discussion and conclusions, which is redundant and reduces the manuscript's coherence.
- The abstract claims are not fully supported by the results, which may mislead readers about the study's outcomes.
- Keywords are not aligned with the content of the manuscript, which may affect its discoverability.
- Figures and tables lack complete captions, making it difficult to interpret their content without referring to the main text.
- Equations are not numbered or defined, which is a critical omission for a technical manuscript.
- Statistical validity is not reported, which is essential for assessing the reliability of the results.
- Reproducibility is not ensured due to the lack of algorithmic detail, making it difficult for independent researchers to replicate the work.
- There is no mention of ethical compliance or conflict-of-interest declaration, which is required for IEEE publications.

Minor Comments:

- The introduction is incomplete and lacks a clear statement of the research problem.
- The methodology section is incomplete and lacks sufficient detail for reproducibility.
- The experiments section is incomplete and lacks a description of the experimental setup.
- The discussion section is incomplete and lacks a thorough analysis of the results.
- The conclusions section is incomplete and lacks a clear summary of the study's findings.

Recommendations:

- Add a literature review section to provide context and establish the novelty of the work.
- Clearly state the research problem in the introduction.
- Explicitly declare the research question or hypothesis.
- Remove the duplicate sections VI and VII and consolidate the discussion and conclusions.
- Ensure that the abstract accurately reflects the results and conclusions of the study.
- Align the keywords with the content of the manuscript to improve discoverability.
- Provide complete captions for all figures and tables.
- Number and define all equations to ensure clarity and reproducibility.
- Report statistical validity, including effect sizes and confidence intervals.
- Include a statement on ethical compliance and conflict-of-interest declaration.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: 95%

Strengths:

- The paper presents a novel physics-informed EANN framework for structural parameter identification under seismic excitation.
- The experimental setup is well-documented, and the comparison with GA and PSO is statistically rigorous.
- The discussion provides a thorough analysis of the results, including statistical inference and physical plausibility.

Weaknesses:

- The introduction lacks a clear problem statement and justification for the proposed method.
- Equation (3) and the matrices A and B are not fully defined, which may hinder reproducibility.
- The methodology section does not clearly describe the implementation of the EANN framework, including the evolutionary search strategy and the integration of physical constraints.

Major Comments:

- The introduction should clearly state the problem and its significance in the context of structural health monitoring and control applications.
- Equation (3) and the matrices A and B need to be fully defined to ensure reproducibility.

Minor Comments:

- The methodology section should provide more details on the implementation of the EANN framework, including the evolutionary search strategy and the integration of physical constraints.

Recommendations:

- Clarify the problem statement and its significance in the introduction.
- Fully define Equation (3) and the matrices A and B in the methodology section.
- Provide more details on the implementation of the EANN framework, including the evolutionary search strategy and the integration of physical constraints.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 85%

Strengths:

- The manuscript correctly applied nonparametric tests (Kruskal-Wallis and Wilcoxon) for comparing stochastic optimization algorithms with non-normal data.
- The use of 30 independent runs was justified for assessing robustness and consistency.
- The coefficient of variation (CV) was used appropriately to assess variability across runs.
- RMSE, MAE, and R^2 were reported and interpreted as measures of accuracy.
- The manuscript included a detailed description of the experimental setup and methodology, which supports reproducibility.
- The use of Pareto fronts and hypervolume indicators was appropriate for multi-objective optimization analysis.

Weaknesses:

- Effect sizes were not reported, which limits the interpretation of statistical significance.
- Bootstrap confidence intervals were mentioned but not quantified or reported in the results.
- Post-hoc Wilcoxon rank-sum tests were not reported, despite the Kruskal-Wallis test being used.
- The manuscript did not report confidence intervals for the metrics (RMSE, MAE, R^2), which is critical for assessing the precision of the results.

Major Comments:

- The manuscript should report effect sizes (e.g., Cohen's d, eta-squared) to complement the reported p-values and provide a more complete picture of the results.
- Bootstrap confidence intervals were mentioned but not quantified or reported in the results. This is a critical omission for assessing the precision of the estimates.
- Post-hoc Wilcoxon rank-sum tests were not reported, despite the Kruskal-Wallis test being used. This is a significant gap in the statistical analysis.

Minor Comments:

- The manuscript should clarify the exact number of resamples used in the bootstrap confidence intervals.
- The manuscript should report confidence intervals for the metrics (RMSE, MAE, R^2) to provide a more complete picture of the results.
- The manuscript should ensure that all statistical tests are clearly labeled and that the p-values are reported with sufficient precision.

Recommendations:

- Report effect sizes (e.g., Cohen's d, eta-squared) to complement the reported p-values.
- Quantify and report bootstrap confidence intervals for all key metrics (RMSE, MAE, R^2).
- Report post-hoc Wilcoxon rank-sum tests to complement the Kruskal-Wallis test results.

- Ensure that all statistical tests are clearly labeled and that the p-values are reported with sufficient precision.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **2.5 / 5.0** | Confidence: 70%

Strengths:

- The manuscript presents a comprehensive framework for structural parameter identification using acceleration-only sensing.
- The methodology is well-defined and includes a state-space formulation and multi-objective optimization.
- The experimental setup is described in detail, including the use of accelerometers and laser sensors.
- The results are discussed in the context of structural health monitoring and adaptive control design.

Weaknesses:

- The figures and tables are not fully described in the text, making it difficult to verify their content and support for the claims made.
- The equations are not fully described in the text, making it difficult to verify their content and support for the claims made.
- The manuscript lacks detailed descriptions of the actual figures and tables, which are essential for reproducibility and validation.
- The manuscript does not provide the actual data from the figures and tables, which is necessary for verifying the claims made.

Major Comments:

- The manuscript lacks detailed descriptions of the actual figures and tables, which are essential for reproducibility and validation.
- The equations are not fully described in the text, making it difficult to verify their content and support for the claims made.
- The manuscript does not provide the actual data from the figures and tables, which is necessary for verifying the claims made.

Minor Comments:

- The captions of some figures are not sufficiently detailed, which reduces their self-explanatory nature.
- The text does not describe the actual content of some tables, which makes it difficult to verify their support for the claims made.
- The equations are not fully described in the text, which makes it difficult to verify their content and support for the claims made.

Recommendations:

- Provide detailed descriptions of the actual figures and tables, including their content and the data they present.
- Fully describe the equations in the text, including their content and the context in which they are used.
- Include the actual data from the figures and tables in the text to support the claims made.

ResultsReviewer

Results validity and empirical support

Score: **4.2 / 5.0** | Confidence: 90%

Strengths:

- The results clearly address the stated objectives of the study, including the comparison of the EANN framework against GA and PSO baselines.
- Quantitative results are well-supported with MSE, RMSE, CV, and Pareto analysis, which are consistent with the abstract and introduction.

- The manuscript includes a comprehensive statistical validation using Kruskal-Wallis, Wilcoxon rank-sum tests, and nonparametric bootstrapping, which aligns with IEEE standards for rigorous experimental validation.
- The results are presented in a structured manner, with clear separation between factual results and interpretation, which is a key strength for clarity and reproducibility.
- The manuscript includes a detailed description of the experimental setup, preprocessing steps, and evaluation metrics, which supports reproducibility.

Weaknesses:

- The results section lacks specific details on the exact values of MSE, RMSE, and CV for each floor and method, which could have been included in tables or figures for clarity and completeness.
- The Pareto front analysis is mentioned but not fully detailed in the results section, making it difficult to assess the depth of the multi-objective optimization trade-offs.
- The hypervolume and spread indicators are referenced but not explicitly reported in the results, which could have provided more insight into the performance of the EANN framework.
- The results section does not clearly describe the exact statistical significance of the differences between the EANN and the baselines, such as p-values or confidence intervals for the reported metrics.
- The results section does not provide a direct comparison of the computational efficiency (e.g., time to convergence) of the EANN framework with the GA and PSO baselines, which is a key metric for algorithmic performance.

Major Comments:

- The results section should include detailed tables or figures that present the exact values of MSE, RMSE, CV, and other metrics for each floor and method to ensure completeness and reproducibility.
- The Pareto front and hypervolume analysis should be expanded with more detailed descriptions and visualizations to support the claims of improved multi-objective trade-offs.
- The statistical significance of the results, such as p-values or confidence intervals, should be explicitly reported to strengthen the validity of the claims.

Minor Comments:

- The results section could benefit from a more detailed description of the 30 independent runs, including how the randomness was controlled and the exact parameters used for each run.
- The results section could include a more explicit comparison of the computational efficiency (e.g., time to convergence) of the EANN framework with the GA and PSO baselines.
- The results section could clarify the exact method used to compute the CV and dispersion metrics for the damping coefficients.

Recommendations:

- Include a table or figure that presents the exact values of MSE, RMSE, CV, and other metrics for each floor and method.
- Expand the Pareto front and hypervolume analysis with more detailed descriptions and visualizations.
- Report the statistical significance of the results, including p-values or confidence intervals for the reported metrics.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.2 / 5.0** | Confidence: 95%

Strengths:

- The Discussion section effectively interprets results in the context of structural identification and control systems, particularly highlighting the advantages of EANN over GA and PSO.
- The Conclusions section clearly states the contribution of the EANN framework to structural health monitoring and adaptive control.
- The manuscript provides a clear experimental validation with statistical significance and reproducibility through multiple runs.

Weaknesses:

- The Discussion section lacks a comprehensive comparison with prior work in the field of structural identification and evolutionary algorithms, which is critical for establishing novelty and relevance.
- The manuscript does not address the limitations of the study, such as computational cost, potential overfitting, or generalizability to other structures or excitation types.
- The Discussion does not sufficiently explain unexpected findings, such as the higher computational cost of EANN despite its improved robustness.
- The Conclusions section does not provide specific, quantified future work directions, which is essential for guiding further research.
- The Discussion overgeneralizes the results by implying broader applicability without sufficient evidence or discussion of domain-specific constraints.

Major Comments:

- The Discussion section fails to compare the proposed EANN framework with prior IEEE publications or state-of-the-art methods in structural identification, which is critical for establishing novelty and relevance.
- The manuscript does not address the limitations of the study, such as computational cost, which is a significant drawback for real-time applications.
- The Discussion does not sufficiently explain the trade-offs between EANN's improved robustness and its higher computational cost, which is a key consideration for practical deployment.

Minor Comments:

- The Discussion could benefit from a more detailed explanation of how the EANN framework improves upon classical metaheuristic approaches in terms of both accuracy and efficiency.
- The Conclusions section could be more specific in quantifying the benefits of the EANN framework, such as the exact percentage improvement in accuracy or reduction in computational time.

Recommendations:

- Include a detailed comparison with prior IEEE publications and state-of-the-art methods in the Discussion section to better contextualize the novelty of the EANN framework.
- Address the limitations of the study, such as computational cost and generalizability, in the Discussion section.
- Provide more detailed explanations of the trade-offs between EANN's robustness and computational cost, and how these factors impact real-world applications.
- Specify future work directions in the Conclusions section, such as exploring hybrid models, reducing computational overhead, or testing on different structural systems.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.0 / 5.0** | Confidence: 55%

Strengths:

- The manuscript contains enough extracted text for a representative writing review.
- Technical terminology and quantitative reporting signals are present in the extracted text.

Weaknesses:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Minor Comments:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Recommendations:

- Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
- Ensure technical terms are defined on first use and used consistently across sections.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 55%

Weaknesses:

- Could not parse model response — raw output stored

Minor Comments:

- The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Recommendations:

- Re-run this agent or check model output
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 95%

Strengths:

- The methodology section provides a clear formulation of the problem and approach, which is a strength in terms of technical clarity.
- The discussion section includes comparative results and visualizations (figures) that support the claims made in the paper.

Weaknesses:

- The manuscript lacks a comprehensive limitations section, which is critical for a Q1 journal publication.
- The limitations are not explicitly stated or discussed, which undermines the credibility of the conclusions.
- The manuscript does not provide sufficient detail on the experimental setup, data collection, or code availability, which hinders reproducibility.
- The manuscript does not address the ethical considerations required for research involving human or animal subjects, if applicable.
- The manuscript does not disclose funding sources, which is a standard requirement for transparency.

Major Comments:

- The manuscript fails to address ethical compliance, including ethical approval, informed consent, and conflict of interest, which are essential for research involving human or animal subjects.
- The manuscript does not provide a data availability statement, which is a fundamental requirement for transparency and reproducibility.
- The manuscript lacks a detailed limitations section, which is critical for a Q1 journal publication and for assessing the validity of the conclusions.
- The manuscript does not disclose funding sources, which is a standard requirement for transparency and accountability.

Minor Comments:

- The methodology section is somewhat vague and lacks sufficient detail for independent replication.
- The discussion section includes some visualizations but does not clearly explain the significance of the results in the context of the limitations.

Recommendations:

- The authors should provide a detailed limitations section that explicitly states the limitations of the study, including generalizability, data constraints, and methodological limitations.
- The authors should include a data availability statement and provide access to the code and raw data used in the study.
- The authors should disclose any potential conflicts of interest and provide information on the funding sources.

- The authors should obtain ethical approval for any research involving human or animal subjects and include this in the manuscript.
- The authors should ensure that the methodology is described with sufficient detail to allow for independent replication of the study.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript is missing a literature review section, which is critical for establishing the context and novelty of the work.
- [2] The research problem is not clearly stated in the introduction, making it difficult to assess the significance of the work.
- [3] The research question or hypothesis is not explicitly declared, which undermines the clarity of the study's objectives.
- [4] Sections VI and VII are duplicates of the discussion and conclusions, which is redundant and reduces the manuscript's coherence.
- [5] The abstract claims are not fully supported by the results, which may mislead readers about the study's outcomes.
- [6] Keywords are not aligned with the content of the manuscript, which may affect its discoverability.
- [7] Figures and tables lack complete captions, making it difficult to interpret their content without referring to the main text.
- [8] Equations are not numbered or defined, which is a critical omission for a technical manuscript.
- [9] Statistical validity is not reported, which is essential for assessing the reliability of the results.
- [10] Reproducibility is not ensured due to the lack of algorithmic detail, making it difficult for independent researchers to replicate the work.
- [11] There is no mention of ethical compliance or conflict-of-interest declaration, which is required for IEEE publications.
- [12] The introduction should clearly state the problem and its significance in the context of structural health monitoring and control applications.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [TITLE] The title is slightly long but remains informative and specific. It could be slightly shortened for improved readability without losing clarity.
- [2] The introduction is incomplete and lacks a clear statement of the research problem.
- [3] The methodology section is incomplete and lacks sufficient detail for reproducibility.
- [4] The experiments section is incomplete and lacks a description of the experimental setup.
- [5] The discussion section is incomplete and lacks a thorough analysis of the results.
- [6] The conclusions section is incomplete and lacks a clear summary of the study's findings.
- [7] The methodology section should provide more details on the implementation of the EANN framework, including the evolutionary search strategy and the integration of physical constraints.
- [8] The manuscript should clarify the exact number of resamples used in the bootstrap confidence intervals.
- [9] The manuscript should report confidence intervals for the metrics (RMSE, MAE, R^2) to provide a more complete picture of the results.
- [10] The manuscript should ensure that all statistical tests are clearly labeled and that the p-values are reported with sufficient precision.
- [11] The captions of some figures are not sufficiently detailed, which reduces their self-explanatory nature.
- [12] The text does not describe the actual content of some tables, which makes it difficult to verify their support for the claims made.

Specific Improvement Recommendations

1. [TITLE] Consider shortening the title slightly to improve readability while maintaining specificity, e.g., 'Physics-Informed EANN for Structural Parameter Identification via Accelerometer Data under Multi-Objective Optimization'.
2. Add a literature review section to provide context and establish the novelty of the work.
3. Clearly state the research problem in the introduction.
4. Explicitly declare the research question or hypothesis.
5. Remove the duplicate sections VI and VII and consolidate the discussion and conclusions.
6. Ensure that the abstract accurately reflects the results and conclusions of the study.
7. Align the keywords with the content of the manuscript to improve discoverability.
8. Provide complete captions for all figures and tables.
9. Number and define all equations to ensure clarity and reproducibility.
10. Report statistical validity, including effect sizes and confidence intervals.
11. Include a statement on ethical compliance and conflict-of-interest declaration.
12. Clarify the problem statement and its significance in the introduction.

This review was generated automatically by the Multi-Agent Paper Review System using Phi-3 Mini (Microsoft) running locally via Ollama. No manuscript content was transmitted to external servers. Rubrics based on Elsevier, IEEE, and Emerald Q1 peer review standards. This document does not constitute an official editorial decision.